

Amey Jotiba Managute

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PROFILE

AI Engineer with an M.S. in Artificial Intelligence and production experience shipping LLM powered systems: AI agents backed by custom MCP tool servers, multimodal RAG pipelines, and scalable Python backends on Azure and AWS.

EDUCATION

University at Buffalo - The State University of New York
Master of Science - Artificial Intelligence, GPA: 3.33/4.0

Buffalo, USA
Aug. 2024 – Feb 2026

University of Mumbai
Bachelor of Engineering - Artificial Intelligence and Data Science, CGPA: 8.44/10

Mumbai, India
Jul 2020 – Jun 2024

EXPERIENCE

AI Engineer | *ProMazo (Client: Dynatrace)*

Feb 2026 – Present | *Remote, USA*

- Built an AI agent in Microsoft Copilot Studio that automates document workflows across company research, Excel model computation (OpenPyXL), and PPTX rendering, powered by a custom OAuth 2.0 secured MCP tool server architected in Python, cutting analyst turnaround from hours to minutes.
- Engineered a stateless, horizontally scalable backend using Azure Redis for session state and Azure Files for artifact staging across replicas, with CI/CD via Azure DevOps and Azure Container Jobs automating TTL based artifact cleanup to control production storage costs.

AI Engineer Intern | *FutureHouse.ai*

Feb 2026 – Jun 2026 | *Remote, USA*

- Built a multimodal AI image generation engine on Google Gemini Flash Image supporting up to 5 reference images with face identity preservation, background removal, and platform aware output, resolving a production Postgres timeout by offloading assets to Supabase Storage.
- Engineered 4 LLM generation pipelines on Google Gemini behind a shared orchestration layer with multi stage JSON repair, and designed multi tenant auth (OAuth, just in time workspace provisioning, magic link verification) with row level security for tenant isolation.

Machine Learning Intern | *The Tann Mann Gaadi*

Jun 2023 – Aug 2023 | *Remote, India*

- Built a natural language query interface for 50+ nontechnical users, cutting average query time by 40%, and automated ETL and feature engineering pipelines with LangChain and PandasAI, reducing data processing time by 60%.

PROJECTS

Multimodal RAG (Retrieval-Augmented Generation) | *Python, FastAPI, Pinecone*

[Demo](#) | [GitHub](#)

- Built a multimodal RAG pipeline extracting text and figures from PDFs, using a vision language model (LLaMA-4 Scout) to summarize figures, with a dual index Pinecone design and intent aware routing that improved answer relevance on mixed visual and text queries.
- Integrated Jina CLIP v2 embeddings with a FastAPI backend and S3 based ingestion, delivering an end to end system for document ingestion, indexing, and grounded question answering.

Manim MCP Server | *Python, FastMCP, Pydantic, Manim*

[GitHub](#)

- Built a Model Context Protocol server letting LLM clients generate 3Blue1Brown style math animations from natural language, with a 5 tool API cutting workflows from 30 to 50 granular operations down to 2 to 3 calls.
- Implemented segment based animation composition and a video rendering pipeline with Pydantic based configuration validation and modular code assembly.

TECHNICAL SKILLS

Core Languages: Python, C++, SQL

ML & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

GenAI & LLMs: RAG, LangChain, Hugging Face, Prompt Engineering, MCP, OpenAI API, Anthropic API, Groq

Backend & APIs: FastAPI, REST APIs, Pydantic, Vector Databases (Pinecone), Redis

MLOps & DevOps: MLflow, Docker, LangSmith, CI/CD, Git, Linux

Cloud (AWS, Azure): S3, EC2, ECS, Lambda, Bedrock, Azure Container Apps, Azure Redis, Azure DevOps